

- 16** Pyruvic acid, $\text{CH}_3\text{COCO}_2\text{H}$, and acetylsalicylic acid, $\text{C}_6\text{H}_5\text{COOH}$, are weak acids. Both acids dissociate in aqueous solutions as follows, where α is the degree of dissociation.



The dissociation constants, K_a of $\text{CH}_3\text{COCO}_2\text{H}$ and K_a of $\text{C}_6\text{H}_5\text{COOH}$, are given in the table below.

	$\text{CH}_3\text{COCO}_2\text{H}$	$\text{C}_6\text{H}_5\text{COOH}$
Dissociation constant/ mol dm^{-3}	1.4×10^{-4}	3.4×10^{-4}

Which statement is correct?

- 1 The pH of 1 mol dm^{-3} of $\text{CH}_3\text{COCO}_2\text{H}$ is lower than that of 1 mol dm^{-3} of $\text{C}_6\text{H}_5\text{COOH}$.
- 2 The value of dissociation constant in terms of initial concentration C and degree of dissociation is $K_a = \frac{\alpha^2}{C(1-\alpha)}$.
- 3 The $\text{p}K_b$ of $\text{CH}_3\text{COCO}_2^-$ is smaller than that of $\text{C}_6\text{H}_5\text{COO}^-$.

A 1 and 2 only **B** 1 only **C** 2 and 3 only **D** 3 only